

Parallels-Utils

Compress, publish, and reuse your Parallels VM images across every machine.

Summary

Parallels-Utils is a Server Factory toolkit for managing Parallels (macOS) virtual-machine images: compressing and synchronizing image “matrices” used for development and testing, publishing them to a remote endpoint, and retrieving/running them on multiple workstations or servers. Usable standalone or as part of Server Factory.

Short description

A Shell/Python toolkit for Parallels VM image lifecycle on macOS. It compresses and synchronizes Parallels images, publishes them to a remote endpoint, and retrieves and runs them across multiple computers — driven by simple settings files, usable independently or within Server Factory.

Long description

Parallels-Utils solves a practical DevOps problem for macOS-based development: teams build “matrices” of Parallels virtual machines (different OSes/configurations used for development and testing), and those images need to be compressed, published, retrieved, and run consistently across many machines. The toolkit provides exactly that lifecycle. A synchronization mechanism compresses Parallels images and keeps them in sync; a publishing mechanism uploads images to a remote endpoint; and a retrieval mechanism lets any workstation or server pull the published images and run them as VMs. Configuration is intentionally simple and file-driven: `image_location.settings` defines where images live on the filesystem, `image_provider.settings` defines the base URL for published images, and `image_sync.sh` defines the upload script — with examples shipped in an `Examples` directory. Operators use `publish_images.sh` to publish and `run.sh` to launch VMs. It requires

Parallels for the appropriate macOS version and Python 3. The toolkit is designed to be dual-use: it can operate as part of a larger Server Factory project or completely independently, reflecting the org's decoupling philosophy. It even ships a short video tutorial link. As part of the Server-Factory family, it complements Qemu-Utills (the Linux/QEMU equivalent), giving the ecosystem VM-image management on both macOS/Parallels and cross-platform/QEMU backends.

Why we built it

Sharing consistent VM development/test environments across a team is tedious — images are large, and every machine needs the same matrix. Parallels-Utills automates compression, publishing, and retrieval so a canonical set of Parallels VMs is reproducible everywhere.

Why it's a game-changer

It turns heavyweight, unwieldy Parallels images into a publishable, syncable artifact set that any machine can pull and run — so a canonical dev/test environment stops being something each engineer rebuilds by hand and becomes something you fetch. It does it with trivial settings-file configuration and zero dependency on the rest of Server Factory, staying true to the org's decoupling philosophy: useful on its own, a good citizen in the larger toolchain.

What's innovative

- Compression + synchronization of Parallels image “matrices” for dev/test.
- Publish/retrieve workflow so images are reusable across many computers.
- Settings-file-driven configuration (location/provider/sync) with shipped examples.
- Dual-use: standalone or as a Server Factory component.

Challenges & solutions

- **Large image distribution:** solved with compression plus a publish-to-remote-endpoint + retrieve workflow.
- **Reproducibility across machines:** solved with provider/location settings so every host resolves the same image set.

- **Ease of use:** solved with simple `publish_images.sh / run.sh` scripts and example settings files.

Tech stack (why + how)

- **Shell** — the publish/run/sync scripts (primary language, ~5.3K bytes).
- **Python 3** — supporting tooling (required dependency, ~3K bytes).
- **Parallels (macOS)** — the virtualization backend being managed.
- **Settings files (.settings)** — declarative configuration for location/provider/sync.

Note: GitHub marks the repository as a fork within the Server-Factory org. Niche, macOS-specific. Not AI-related.